

Investigate the stability on the mesh size (issue #19)

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1 Short summary

This report checks the mesh sensitivity of the 3-phase machine (synrm_3f_36s). The coil, stator, air-gap band and rotor mesh sizes are varied separately to verify suitable settings for the torque calculation.

2 Method

The coil, stator and air-gap band meshes are controlled in Ansys by a prescribed maximum element length, while the rotor mesh is controlled by the rotor mesh length passed to the computation. The air-gap band corresponds to the Band region in the Ansys model. Four separate mesh sweeps are performed for the coil, stator, air-gap band and rotor regions. In each sweep, only one region is varied from 0.1 to 3.0 mm in steps of 0.1 mm, while the other three regions are fixed at 0.5 mm.

Two fixed Pareto optimal rotor designs were tested, one from HacklGenerator_SixLambdas and one from HacklGenerator_3BrokenLines. TorAvg and TorRippleRms were evaluated with `analyze_results()`. The sweep is implemented in `scripts/sensitivity_mesh/sensitivity_mesh.py`.

3 Results

Figures 1 and 2 show the complete mesh sweep for the four regions. The coil changes only slightly over most of the range. The stator, air-gap band and rotor results show more noticeable fluctuations as the mesh size is varied, especially for the 3BrokenLines design.

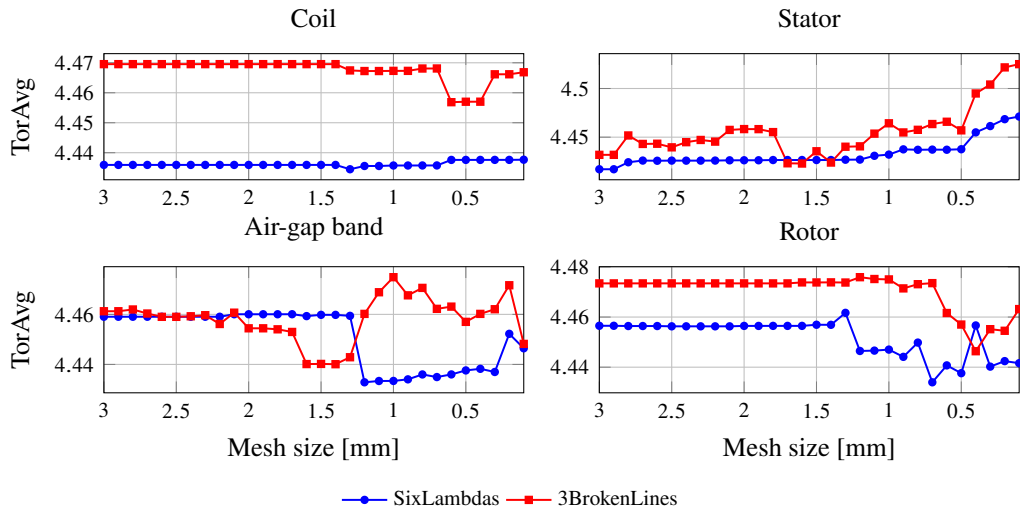


Figure 1: Mean torque as the mesh size is varied in each region.

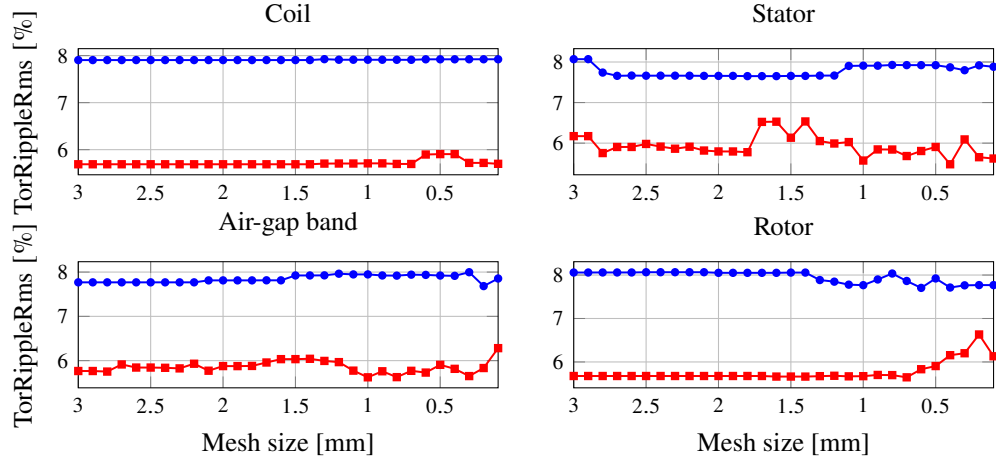


Figure 2: Torque ripple as the mesh size is varied in each region.

4 Stability analysis

The 0.5 mm case is used as the reference. At a stator mesh size of 0.6 mm, the change in TorAvg is 0.01% for SixLambdas and 0.20% for 3BrokenLines. The corresponding changes in TorRippleRms are 0.00 and 0.10 percentage points.

The coil is less sensitive to the mesh size. At 3.0 mm, the change in TorAvg is 0.04% for SixLambdas and 0.28% for 3BrokenLines. The ripple changes by 0.02 and 0.22 percentage points.

The air-gap band and rotor results show larger fluctuations as the mesh is coarsened. Their 0.5 mm settings are retained.

5 Recommendation

For `synrm_3f_36s`, the recommended maximum mesh lengths are

$$h_{\text{rotor}} = 0.5 \text{ mm}, \quad h_{\text{airgap}} = 0.5 \text{ mm}, \quad h_{\text{stator}} = 0.6 \text{ mm}, \quad h_{\text{coil}} = 3.0 \text{ mm}.$$